

Online Appendix

Relative Earnings and Labor Force Exit Rate: Evidence from US Prime-Age Men

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Appendix

A.1 A Technical Note on Labor Force Participation and Exit Rate

Rearranging Equation 2.1, the labor force non-participation rate, defined as the share of the population not in the labor force, can be written as a function of the population-level labor force exit rate ε_t^* and its lagged terms, each weighted by the survival function $s(\tau)$ defined in Section 2:

$$NLFP R_t = \sum_{k=0}^t \varepsilon_{t-k}^* \cdot s(k) \quad (\text{A1})$$

As in Equation 2.1, each term in the summation measures an exit flow relative to the population, so the terms share a common base and sum to the share of the population outside the labor force at t .

The equation implies that changes in either the survival function or the exit rate can move the non-participation rate. Empirically, however, $s(\tau)$ has been largely stable over the last 40 years. Figure A1 plots the labor force exit survival function based on the linked IPUMS-CPS basic monthly data series from 1980 through 2019. The function shifts slightly between 2010 and 2019, likely reflecting worker movement after the Great Recession, but otherwise exhibits minimal changes over time. Changes in the labor force exit rate therefore drive the changes in prime-age men's non-participation rate, an observation also reported by Elsbey et al. (2019).

Assume hereafter that $s(\tau)$ is time invariant. The change in the non-participation rate from period t' to t , where $T < t' < t$, can be written as:¹

$$\begin{aligned} \Delta NLFP R_{t',t} &= \sum_{k=0}^{t'} (\varepsilon_{t-k}^* - \varepsilon_{t'-k}^*) \cdot s(k) + \sum_{k'=t'+1}^t \varepsilon_{t-k'}^* \cdot s(k') \\ &= \sum_{k=0}^{t'} (\varepsilon_{t-k}^* - \varepsilon_{t'-k}^*) \cdot s(k) \end{aligned} \quad (\text{A2})$$

¹The second part of equation A2 can be dropped because of the assumption that $T < t' < t$ so that $s(\tau) \rightarrow 0 \forall \tau > t'$.



Figure A1: Labor Force Exit Survival Function, 1980–2019

Source(s): IPUMS-CPS 1980-2019.

Note(s): The figure plots the labor force exit survival function using the linked IPUMS-CPS basic monthly data series. The sample consists of respondents who had a recorded labor force exit event between month-in-sample 1 and 2. The survey design of the Current Population Survey does not allow the observation of the employment transitions between the 5th and the 12th month during the 16-month survey period, which is the reason the segment between month 2 and month 11 appears flat.

In other words, the change in $NLFPR$ from t' to t depends on $\Delta\varepsilon_{t',t}^*$, the changes in the population-level exit rate from t' to t , and the changes in the population-level exit rate in previous periods, $\Delta\varepsilon_{t'-k,t-k}^*$. An increase in the exit rate hence has both a contemporary effect on the current non-participation rate and lagged effects on the future non-participation rates from workers staying outside of the labor force past the current period.

Equation A1 also indicates that the percentage change in the non-participation rate directly relates to the percentage change in the population-level exit rate. The percentage change in the non-participation rate from period t' to t , where $T < t' < t$, can be derived as:

$$\% \Delta NLFPR_{t',t} = \frac{\sum_{k=0}^{t'} (\varepsilon_{t-k}^* - \varepsilon_{t'-k}^*) \cdot s(k)}{\sum_{k=0}^{t'} \varepsilon_{t'-k}^* \cdot s(k)} \quad (\text{A3})$$

Under the assumption that the exit rate grows at a fixed rate r , such that $\varepsilon_{t+k}^* = \varepsilon_t^* (1+r)^k$, the percentage change in the non-participation rate is equal to the percentage change in the exit rate:

$$\begin{aligned} \% \Delta NLFPR_{t',t} &= \frac{\sum_{k=0}^{t'} (\varepsilon_{t-k}^* - \varepsilon_{t'-k}^*) \cdot s(k)}{\sum_{k=0}^{t'} \varepsilon_{t'-k}^* \cdot s(k)} \\ &= \frac{\sum_{k=0}^{t'} \varepsilon_{t'-k}^* [(1+r)^{t-t'} - 1] \cdot s(k)}{\sum_{k=0}^{t'} \varepsilon_{t'-k}^* \cdot s(k)} \\ &= (1+r)^{t-t'} - 1 \\ &= \% \Delta \varepsilon_{t',t}^* \end{aligned} \quad (\text{A4})$$

In the main analysis, the labor force exit rate, ε_t , is defined as the share of the *labor force*, rather than the share of the population, exiting the labor force at t , such that

$$\varepsilon_t = \frac{\varepsilon_t^*}{LFPR_t}.$$

The two measures differ by a factor of $1/LFPR_t$, which rises as participation falls. Because the participation rate moves slowly relative to the exit rate, the two series follow similar growth paths for most of the study period, and both grow alongside the non-participation

rate (Figure 2.1). Between 1980 and 2019, the labor force exit rate grew by 84 percent and the population-level exit rate by 74 percent, while the non-participation rate grew by 87 percent. The gap between the two exit measures stays small through the mid-2000s and widens after the Great Recession, when the decline in prime-age men’s labor force participation accelerated and the scaling factor rose more quickly.

A.2 Relationship Between State Characteristics and IV Exposure Shares

In the 2SLS analysis, the reference group earnings are instrumented by the inner product between a state’s initial industry shares in the period between 1976 and 1979 and the industry-level wage rates in their region, defined by census divisions. This instrument can be viewed as a special case of standard shift-share instruments. In this case, the exposure shares of the instrument are a state’s initial region-industry shares, defined as the interaction between a state’s initial industry shares and region dummies and the earnings shocks are common within each region-industry in each period. The instrument is therefore analogous to instrumenting using a state’s initial region-industry shares and their interactions with time period dummies. Accordingly, this reconfiguration allows the adoption of test techniques derived by Goldsmith-Pinkham et al. (2020) to identify the key regions or industries associated with the largest Rotemberg weights, which measure their relative importance in determining the 2SLS estimator.

If states in these key regions or with high initial shares of the key industries exhibit other baseline characteristics that predict more substantial growth in labor force exit rate, it would suggest that the exposure shares are not exogenous to changes in the exit rate. To test this, I follow the approach suggested by Goldsmith-Pinkham et al. (2020) and regress the initial industry shares (or region indicators) on selected 1976–1979 state-level characteristics that could predispose the states to experience more substantial increases in prime-age male labor force exit rate. These characteristics include the share of prime-age men without a four-year

college degree, not married, non-Hispanic Black, or employed in sales and administrative support occupations.²

For computational feasibility, I adopt a modified sample and model specification for this exercise. The sample is collapsed from individual-level to state-by-skill level observations, spanning the years 1980, 1985, 1990, 1995, 2000, 2005, 2010, 2015, and 2019. In this simplified model, the only endogenous variable in the model is the reference group earnings, with the expected earnings variable included as an exogenous covariate. The instrument for the reference group earnings is constructed with a more parsimonious industry classification scheme of 35 industries. Despite the modification, the OLS and 2SLS results of the original and modified models remain broadly comparable (Table A1) and this consistency allows me to draw insights from the simplified model to inform the primary analysis.

Table A2 presents the test results. In panel A, the dependent variables are a state's initial shares of the top four industries with the largest Rotemberg weights. In panel B, the dependent variables are indicator variables of the top four census divisions. Across the set of key industries and regions, there appear to be limited correlations between the exposure shares and these state characteristics. In a few cases, negative correlations are observed, suggesting that the direction of bias, if any, would likely attenuate the 2SLS estimator towards zero. The only exposure share positively associated with a factor predicting larger exit growth is the indicator for the South Atlantic census division, which is correlated with the initial share of non-Hispanic Black men. Excluding states in the South Atlantic census division from the sample, however, has limited impact on the OLS and 2SLS estimators. These findings provide supportive evidence for the exogeneity of the exposure shares of the instrument.

²These four factors are selected from a broader set of 11 demographic characteristics spanning race/ethnicity, marital status, educational attainment, and occupations. An OLS regression analysis indicates that this subset of characteristics is associated with significantly higher growth trends in the prime-age male labor force exit rate.

Table A1: Results from Original and Modified Models

	Original		Modified	
	(1)	(2)	(3)	(4)
Expected earnings	-1.23*** (0.06)	-1.38*** (0.08)	-1.26*** (0.14)	-1.29*** (0.14)
Reference group earnings	1.18*** (0.26)	1.99* (0.88)	1.53** (0.35)	2.21** (0.74)
Method	OLS	2SLS	OLS	2SLS
Observations	7,903,696	7,903,696	4,583	4,583
R^2	0.04	0.04	0.57	0.53
$\hat{\gamma}$	0.96	1.45	1.22	1.71
First-stage Kleibergen-Paap rk Wald F statistic		11.69		59.40

Source(s): IPUMS-CPS 1980–2019.

Note(s): The dependent variable of the regressions is the probability of labor force exit measured in % pts. The sample consists of US prime-age men for the 1980–2019 period. All models include state- and individual-level controls as well as state and year fixed effects. The results are weighted by the CPS longitudinal weight for linking individuals between adjacent months. Standard errors are clustered at the state level.

† for $P < 0.10$, * for $P < 0.05$, ** for $P < 0.01$, *** for $P < 0.001$.

Table A2: Relationship Between Instrument Shifters and State Characteristics

	A. Top Four Industries			
	(1) Professional Services	(2) Retail Trade	(3) Construction	(4) Agriculture
Non-college	-0.31*** (0.06)	0.01 (0.06)	-0.11* (0.06)	-0.14 (0.12)
Non-married	0.02 (0.06)	0.04 (0.06)	-0.07 (0.06)	-0.15 (0.12)
Non-Hispanic Black	0.03 (0.03)	-0.05 [†] (0.03)	0.02 (0.03)	-0.09 [†] (0.05)
Sales and administration	-0.08 (0.16)	-0.07 (0.17)	-0.78*** (0.15)	-0.97** (0.32)
Observations	51	51	51	51
R^2	0.52	0.09	0.41	0.31
	B. Top Four Census Divisions			
	(5) New England	(6) West North Central	(7) South Atlantic	(8) East South Central
Non-college	-1.49 (1.19)	-1.54 (1.36)	-1.14 (1.46)	1.31 (1.07)
Non-married	0.04 (1.25)	-1.90 (1.43)	-1.42 (1.54)	-0.79 (1.13)
Non-Hispanic Black	-0.65 (0.55)	-0.77 (0.63)	3.06*** (0.68)	0.82 (0.49)
Sales and administration	-3.27 (3.25)	-2.45 (3.73)	-4.28 (4.02)	-2.15 (2.94)
Observations	51	51	51	51
R^2	0.10	0.11	0.32	0.20

Source(s): IPUMS-CPS 1980–2019.

Note(s): The dependent variable is a state's initial 1976–1979 employment share in the indicated industry (panel A) or an indicator for the indicated census division (panel B). The four industries and four divisions shown are those with the largest Rotemberg weights. The explanatory variables are 1976–1979 state characteristics. The sample consists of the 50 US states and the District of Columbia. The results are weighted by the male population in each state.

[†] for $P < 0.10$, * for $P < 0.05$, ** for $P < 0.01$, *** for $P < 0.001$.

A.3 Results by CPS Month-in-sample

Because of the rotating panel design of the CPS, a worker can appear in this study’s primary analysis sample multiple times in their second to fourth and sixth to eighth month-in-sample (MIS 2 through 4 and MIS 6 through 8). This feature raises concerns about estimation bias when unobserved individual characteristics are correlated with the outcome and independent variables. Conceptually, the problem can be addressed by restricting the sample to a single MIS group so that each worker is included in the sample no more than once during their CPS survey window. In practice, survey methodologists have found that survey responses to CPS labor force participation questions vary non-randomly across CPS MIS groups due to panel conditioning effects (Warren and Halpern-Manners, 2012). It is unclear whether using a single MIS introduces additional sources of bias.

As a robustness test to assess whether using a single MIS group as well as the choice of MIS group affects the estimation results, I estimate the baseline model separately for each of the six MIS groups. Table A3 reports the results. The point estimates based on the sub-samples of MIS 2, 3, 6, and 7 are similar and qualitatively comparable to the baseline results in Table 6.1. By contrast, the point estimates are smaller based on the sub-samples of MIS 4 and 8. The lower marginal effects likely reflect the generally lower exit rate reported by these two MIS groups.

The difference shown in MIS 4 and 8 samples is consistent with the observation in Warren and Halpern-Manners (2012) that CPS survey respondents are disinclined to repeat socially stigmatized responses in consecutive surveys, which results in lower unemployment estimates based on MIS 4 or 8 compared with other months in survey. Aside from the anomaly in results based on MIS 4 and 8, there is no strong evidence that using a pooled sample qualitatively affects the estimation results.

Table A3: Estimation Results by CPS Month in Survey (MIS)

	(1)	(2)	(3)	(4)	(5)	(6)
Expected earnings	-1.56*** (0.10)	-1.36*** (0.11)	-0.96*** (0.11)	-1.33*** (0.08)	-1.20*** (0.09)	-0.99*** (0.10)
Reference group earnings	1.64** (0.47)	1.39*** (0.35)	0.96* (0.46)	1.46*** (0.39)	1.19** (0.40)	0.42 (0.44)
MIS	2nd	3rd	4th	6th	7th	8th
Observations	1,296,172	1,328,726	1,331,211	1,302,716	1,321,230	1,323,641
R^2	0.05	0.05	0.04	0.05	0.04	0.04
$\hat{\gamma}$	1.05	1.03	1.00	1.10	0.99	0.43
Average exit rate (%)	1.88	1.65	1.51	1.76	1.54	1.44

Source(s): IPUMS-CPS 1980–2019.

Note(s): The dependent variable of the regression is the probability of labor force exit measured in % pts. The sample consists of US prime-age men for the 1980–2019 period. All models include state- and individual-level controls as well as state and year fixed effects. The results are weighted by the CPS longitudinal weight for linking individuals between adjacent months. Standard errors are clustered at the state level.

† for $P < 0.10$, * for $P < 0.05$, ** for $P < 0.01$, *** for $P < 0.001$.

A.4 Pathways to Labor Force Exit

This section describes the employment and income pathways through which prime-age men transition from active labor force participation to both long-term and short-term withdrawal from the workforce. The analysis leverages longitudinal data from the Survey of Income and Program Participation (SIPP) 2008 panel to gain insights into this dynamic process. The dataset comprises US prime-age men who were in the labor force during one or more months of the SIPP 2008 sample period.

Conditional on being in the labor force in month t , the observations are grouped based on the individuals' labor force exit decision in the subsequent month $t + 1$ and their labor force participation status in months $t + 2$ through $t + 13$. This categorization results in three mutually exclusive groups: long-term dropouts, short-term dropouts, and stayers. The long-term dropouts are workers who left the labor force in $t + 1$ and remained outside the workforce for more than 12 months. The short-term dropouts are workers who exited the labor force in $t + 1$ but returned within 12 months. Finally, the stayers are individuals who remained in the labor force in $t + 1$.

For each of the three groups, I summarize their employment patterns, personal income, transfer benefits, and health status in the 12 months preceding the labor force exit event ($t - 11$ through t), as well as the 12 months following the exit event ($t + 2$ through $t + 13$). This provides an illustration of the trajectories leading up to and resulting from a prime-age man's decision to withdraw from the labor force.

Table A4 reports the results. Columns 1 and 2 in Panel A summarize the employment activities of long-term and short-term dropouts, respectively. On average, long-term dropouts spent 27.5 percent of their time employed full-time in the year preceding the exit event, while short-term dropouts spent 38.8 percent of their time in full-time work during this period. The remainder of their time was divided into part-time employment (19.5 percent and 19.6 percent), job search (31.2 percent and 30.3 percent), and short-term exit from the labor force (24.6 percent and 13.0 percent). By contrast, workers who stayed in the labor force

dedicated approximately 80 percent of their time to full-time employment, with significantly less time spent job searching or temporarily non-participating. These differences highlight that labor force exit, particularly long-term exit, often follows an extended period of weak attachment to the workforce.

Reflecting their weak attachment to the workforce, both long-term and short-term dropouts had low personal incomes in the 12 months preceding the exit event (Panel B). The opportunity cost of non-participation therefore tends to be smaller for the dropouts compared to the stayers. For some long-term dropouts, their lost earnings were partially replaced by Social Security benefits following the exit event, as indicated by an 8.3 percentage point increase in the share receiving such benefits after t . There is no strong evidence that the short-term dropouts' decision to exit the labor force was correlated with changes in the availability of transfer benefits. Additionally, the long-term dropouts reported a higher rate of work-limiting disability compared to the other two groups prior to the exit event, and this gap significantly widened after they left the labor force. This increase suggests that the decision to sever ties with the labor force nearly permanently often coincides with self-reported disability for many workers.

These descriptive patterns suggest that workers tend to leave the labor force when the expected returns from participation are small. Lengthy unemployment, low income, and poor physical condition are strong predictors of both a worker's decision to exit the labor force and the duration of their non-participation spell.

Table A4: Prime-age Men's Employment and Income Pathways to Labor Force Exit

	Long-term dropouts	Short-term dropouts	Stayers
	(1)	(2)	(3)
A. Employment			
Percent of time full-time employed			
Months 1-12 before event	27.5	38.8	80.0
Months 1-12 after event	0.0	30.4	80.3
Percent of time part-time employed			
Months 1-12 before event	19.5	19.6	11.6
Months 1-12 after event	0.0	13.8	11.3
Percent of time searching for work			
Months 1-12 before event	31.2	30.3	6.0
Months 1-12 after event	0.0	24.0	5.9
Percent of time out of labor force			
Months 2-12 before event*	24.6	13.0	2.6
Months 1-12 after event	100.0	31.8	2.5
B. Income and Benefits			
Median monthly personal income			
Months 1-12 before event	\$980	\$1,404	\$3,532
Months 1-12 after event	\$109	\$987	\$3,516
Percent receiving Social Security benefits			
Months 1-12 before event	8.7	1.5	0.7
Months 1-12 after event	17.0	2.1	0.8
Percent receiving unemployment benefits			
Months 1-12 before event	6.6	10.5	3.7
Months 1-12 after event	2.9	10.3	3.8
C. Health and Education			
Percent with work-limiting disability			
Months 1-12 before event	38.9	11.6	4.6
Months 1-12 after event	57.8	14.3	5.1

* For "Percent of time out of labor force", the summary statistics reflect labor market activities from month $t - 11$ through $t - 1$ because, by sample definition, a worker has to be in the labor force at t .

Source(s): SIPP 2008 panel.

Note(s): The sample consists of prime-age men who reported participating in the labor force in one or more months from September 2008 through August 2012. The results are weighted by the SIPP sample weight.

A.5 Business Cycles, Relative Earnings, and Prime-age Men’s Labor Force Exit Decision

Whether business cycles influence the degree to which prime-age men consider relative earnings in their labor force exit decision does not have a clear a priori answer. On one hand, unemployed workers likely have higher wage elasticity of labor force exit compared with employed workers, as most of the latter do not actively contemplate leaving the labor force. When more workers become unemployed during a downturn, workers’ overall responsiveness to changes in their relative earnings may increase. On the other hand, workers may place lower weight on the non-pecuniary return on employment when jobs are scarce, which suggests that their sensitivity to their peers’ earnings may decline during a downturn.

To examine these two channels, I augment Equation 4.1 with two sets of interaction terms. The first set interacts a worker’s expected and reference group earnings with the prime-age male unemployment rate in his state of residence, which measures aggregate labor market conditions. The second set interacts the earnings variables with his own unemployment status in the previous calendar month. Table A5 reports the estimation results. All models are estimated by OLS. Both moderators are themselves likely to respond to the latent demand shocks discussed in Section 4.2, which makes the interaction terms endogenous as well. The estimates in this section therefore describe how the observed relationship varies with labor market conditions rather than identify a causal business cycle effect.

Column 2 includes both sets of interaction terms. Conditional on a worker’s own unemployment status, a higher state unemployment rate is associated with a weaker response to both earnings measures, and the attenuation is more pronounced for the reference group earnings. For a worker who was employed in the previous calendar month, the implied discount factor is 0.80 when the state prime-age male unemployment rate is 3 percent, 0.59 when it is 5 percent, and 0.10 when it is 8 percent. Prime-age men thus appear to weigh their earnings against their peers’ less heavily when local labor market conditions are weak.

The individual margin works in the opposite direction. Column 1 shows that workers who

were unemployed in the previous calendar month respond more strongly to both earnings measures than workers who were employed. At a state unemployment rate of 5 percent, their implied discount factor is 1.88, compared with 0.59 for workers who were employed in the previous calendar month.³

Because a downturn both attenuates the estimated response within each group and moves more workers into the more responsive unemployed group, the two patterns offset each other for an average worker. Column 3 illustrates this. When the interaction terms with individual unemployment status are excluded from the model, the state unemployment interaction terms fall in magnitude and are no longer statistically significant. The offset reflects the changing composition of the labor force over the business cycle rather than an absence of a relationship between labor market conditions and workers' sensitivity to relative earnings.

³For unemployed workers, the coefficient on expected earnings is roughly half the size of the coefficient on the reference group earnings, which suggests that unemployed individuals discount the growth in the measured expected earnings when forming their own earnings expectations.

Table A5: Relative Earnings, Business Cycle, and Prime-age Men's Labor Force Exit Rate

	(1)	(2)	(3)
Expected earnings	-0.94*** (0.05)	-1.33*** (0.09)	-1.43*** (0.09)
Reference group earnings	0.61* (0.26)	1.35*** (0.33)	1.36*** (0.34)
Expected earnings $\times$ Prime-age male unemployment rate		0.08*** (0.01)	0.04** (0.01)
Reference group earnings $\times$ Prime-age male unemployment rate		-0.16*** (0.04)	-0.04 (0.05)
Expected earnings $\times$ Unemployed in the previous month	-5.75*** (0.39)	-5.81*** (0.38)	
Reference group earnings $\times$ Unemployed in the previous month	12.04*** (1.81)	12.15*** (1.81)	
Method	OLS	OLS	OLS
Observations	7,903,696	7,903,696	7,903,696
R^2	0.05	0.05	0.04

Source(s): IPUMS-CPS 1980–2019.

Note(s): The dependent variable of the regression is the probability of labor force exit measured in % pts. The sample consists of US prime-age men for the 1980–2019 period. All models include state- and individual-level controls as well as state and year fixed effects. The results are weighted by the CPS longitudinal weight for linking individuals between adjacent months. Standard errors are clustered at the state level.

† for $P < 0.10$, * for $P < 0.05$, ** for $P < 0.01$, *** for $P < 0.001$.

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